

10-27

Exercise

1) Given $M = \begin{bmatrix} 0.8 & 0.5 \\ 0.2 & 0.5 \end{bmatrix}$ find its

diagonalization and compute the steady-state matrix M^∞

$$\begin{aligned} |M - \lambda I| &= \begin{vmatrix} 0.8 - \lambda & 0.5 \\ 0.2 & 0.5 - \lambda \end{vmatrix} \\ &= \lambda^2 - \frac{13}{10}\lambda + \frac{3}{10} \\ &= (\lambda - 1)\left(\lambda - \frac{3}{10}\right) \\ \lambda &= 1, \frac{3}{10} \end{aligned}$$

$$\lambda_1 = 1 : M - I = \begin{bmatrix} -0.2 & 0.5 \\ 0.2 & -0.5 \end{bmatrix}$$

$$\vec{v}_1 = \begin{bmatrix} 0.5 \\ 0.2 \end{bmatrix} \left(= \begin{bmatrix} 5/10 \\ 2/10 \end{bmatrix} \right)$$

$$\lambda_2 = 0.3 : M - 0.3I = \begin{bmatrix} 0.5 & 0.5 \\ 0.2 & 0.2 \end{bmatrix}$$

$$\vec{V}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\text{Thus } M = \overset{S}{\begin{bmatrix} 0.5 & -1 \\ 0.2 & 1 \end{bmatrix}} \overset{\wedge}{\begin{bmatrix} 1 & \\ & 0.3 \end{bmatrix}} \overset{S^{-1}}{\begin{bmatrix} 0.5 & -1 \\ 0.2 & 1 \end{bmatrix}}^{-1}$$

and hence

$$M^\infty = \lim_{k \rightarrow \infty} M^k$$

$$= \lim_{k \rightarrow \infty} S \wedge^k S^{-1}$$

$$= \lim_{k \rightarrow \infty} \begin{bmatrix} 0.5 & -1 \\ 0.2 & 1 \end{bmatrix} \begin{bmatrix} 1 & \\ & 0.3^k \end{bmatrix} \begin{bmatrix} 0.5 & -1 \\ 0.2 & 1 \end{bmatrix}^{-1}$$

$$= \begin{bmatrix} 0.5 & -1 \\ 0.2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0.5 & -1 \\ 0.2 & 1 \end{bmatrix}^{-1}$$

$$= \frac{10}{7} \begin{bmatrix} 0.5 & -1 \\ 0.2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -0.2 & 0.5 \end{bmatrix}$$

$$= \frac{10}{7} \begin{bmatrix} 0.5 & -1 \\ 0.2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

$$= \frac{10}{7} \begin{bmatrix} 0.5 & 0.5 \\ 0.2 & 0.2 \end{bmatrix} = \begin{bmatrix} 5/7 & 5/7 \\ 2/7 & 2/7 \end{bmatrix}$$

Note that for $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ where $x_1 + x_2 = 1$

we have that $M^\infty \vec{x} = \begin{bmatrix} 5/7 & 5/7 \\ 2/7 & 2/7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$$= \begin{bmatrix} \frac{5}{7}(x_1 + x_2) \\ \frac{2}{7}(x_1 + x_2) \end{bmatrix} = \begin{bmatrix} 5/7 \\ 2/7 \end{bmatrix}$$

Interpretation: Given $M = \begin{bmatrix} a & b \\ 1-a & 1-b \end{bmatrix}$ and

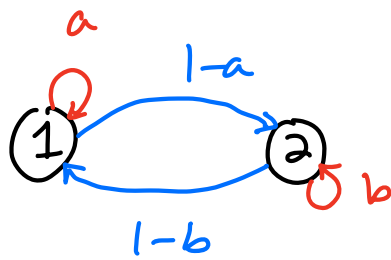
$\vec{x}_t = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}$ representing the "state" (probabilistic)

at time t then $M \vec{x}_t = \vec{x}_{t+1}$ meaning that if $x_1(t)$ is the probability of being in state 1 at time t and $x_2(t)$ is the probability of being in state 2 at time t , then

$$\begin{bmatrix} x_1(t+1) \\ x_2(t+1) \end{bmatrix} = M \vec{x}_t = \begin{bmatrix} ax_1 + bx_2 \\ (1-a)x_1 + (1-b)x_2 \end{bmatrix}$$

represents the change of state where an object in state 1 has a probability of a of staying and a probability of $1-a$ of leaving for

state 2.



Now let $\vec{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$ be such that $v_1 + v_2 = 1$ and $M\vec{v} = \vec{v}$. Then $\vec{v}$ is the steady-state and is stable as for any $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ with $x_1 + x_2 = 1$ we have that

$$\lim_{k \rightarrow \infty} M^k \vec{x} = \lim_{t \rightarrow \infty} x_t = \vec{v}$$

Def 16.1 The equation $\vec{v}_{t+1} = A\vec{v}_t$ is called a linear difference equation.

Linearly Recursive Sequences

Def 16.2 (s_n) is linearly recursive sequence iff

$$s_n = \sum_{i=1}^k c_i s_{n-i}$$

Setting $\vec{x}_n = \begin{bmatrix} s_{n+k-1} \\ \vdots \\ s_{n+1} \\ s_n \end{bmatrix} \in \mathbb{R}^k$ we have that

$$\vec{x}_{n+1} = A \vec{x}_n \quad \text{for some } A \in \mathbb{R}^{k \times k}$$

($k=3$)

Example: Given $s_n = s_{n-1} - 2s_{n-2} + 3s_{n-3}$

Set $\vec{x}_n = \begin{bmatrix} s_{n+2} \\ s_{n+1} \\ s_n \end{bmatrix}$ and let A be such

that $A \vec{x}_n = \vec{x}_{n+1} = \begin{bmatrix} s_{n+3} \\ s_{n+2} \\ s_{n+1} \end{bmatrix}$. We can find

such an $A \in \mathbb{R}^{3 \times 3}$ as follows

$$\begin{bmatrix} s_{n+3} \\ s_{n+2} \\ s_{n+1} \end{bmatrix} = \begin{bmatrix} s_{n+2} - 2s_{n+1} + 3s_n \\ s_{n+2} \\ s_{n+1} \end{bmatrix} = A \begin{bmatrix} s_{n+2} \\ s_{n+1} \\ s_n \end{bmatrix}$$

$$\text{Thus } A = \begin{bmatrix} 1 & -2 & 3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Exercise :

2) Consider the Fibonacci sequence defined by $f_n = f_{n-1} + f_{n-2}$ where $f_1 = f_2 = 1$.

Letting $\vec{x}_n = \begin{bmatrix} f_{n+1} \\ f_n \end{bmatrix}$ find a matrix

A such that $\vec{x}_{n+1} = A\vec{x}_n$ and

show that $f_n = \frac{\varphi^n - \psi^n}{\varphi - \psi}$ where

$\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$. Deduce

that $\lim_{n \rightarrow \infty} \frac{f_{n+1}}{f_n} = \frac{1+\sqrt{5}}{2}$.

$$\text{Let } A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \text{ so that } A \begin{bmatrix} f_{n+1} \\ f_n \end{bmatrix} = \begin{bmatrix} f_{n+1} + f_n \\ f_{n+1} \end{bmatrix} \\ = \begin{bmatrix} f_{n+2} \\ f_{n+1} \end{bmatrix}$$

Then A has e-values of

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 - \lambda - 1$$

$$\lambda = \frac{1 \pm \sqrt{5}}{2}$$

and letting $\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$ we

have that $\varphi + \psi = 1$ and

$$A - \varphi I = \begin{bmatrix} 1-\varphi & 1 \\ 1 & -\varphi \end{bmatrix} = \begin{bmatrix} \psi & 1 \\ 1 & -\varphi \end{bmatrix}$$

$$A - \psi I = \begin{bmatrix} \varphi & 1 \\ 1 & -\psi \end{bmatrix}$$

thus A has e-vectors of $\begin{bmatrix} \varphi \\ 1 \end{bmatrix}$ and $\begin{bmatrix} \psi \\ 1 \end{bmatrix}$

$$\text{Now } A = \begin{bmatrix} \varphi & \psi \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \varphi & \psi \\ 1 & 1 \end{bmatrix}^{-1}$$

$$\text{and as } \begin{bmatrix} f_{n+1} \\ f_n \end{bmatrix} = A^{n-1} \begin{bmatrix} f_2 \\ f_1 \end{bmatrix} = A^{n-1} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

we have that

$$\begin{bmatrix} f_{n+1} \\ f_n \end{bmatrix} = \begin{bmatrix} \varphi & \psi \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \varphi^{n-1} & \psi^{n-1} \end{bmatrix} \begin{bmatrix} \varphi & \psi \\ 1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\varphi - \psi} \begin{bmatrix} \varphi & \psi \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \varphi^{n-1} & \psi^{n-1} \end{bmatrix} \begin{bmatrix} 1 & -\psi \\ -1 & \varphi \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\varphi + \psi = 1$$

$$\varphi = 1 - \psi$$

$$= \frac{1}{\varphi - \psi} \begin{bmatrix} \varphi & \psi \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \varphi^{n-1} & \\ & \psi^{n-1} \end{bmatrix} \begin{bmatrix} \varphi \\ -\psi \end{bmatrix}$$

$$= \frac{1}{\varphi - \psi} \begin{bmatrix} \varphi & \psi \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \varphi^n \\ -\psi^n \end{bmatrix}$$

$$= \frac{1}{\varphi - \psi} \begin{bmatrix} \varphi^{n+1} - \psi^{n+1} \\ \varphi^n - \psi^n \end{bmatrix}$$

Thus $f_n = \frac{\varphi^n - \psi^n}{\varphi - \psi}$ and

$$\lim \frac{f_{n+1}}{f_n} = \frac{\varphi^{n+1} - \psi^{n+1}}{\varphi^n - \psi^n}$$

Now as $|\varphi| > |\psi|$, $\lim \frac{\varphi^{n+1} - \psi^{n+1}}{\varphi^n - \psi^n} = \varphi$

Complex E-values of Matrices over $\mathbb{R}$

Theorem 16.3 Let $p(z) = \sum_{i=0}^n a_i z^i$ where $a_i \in \mathbb{R}$. Then $p(\bar{z}) = \overline{p(z)}$

The proof uses the following properties

$$i. \quad \overline{z+w} = \overline{z} + \overline{w}$$

$$ii. \quad \overline{\overline{z}} = z$$

$$iii. \quad \overline{z} = z \text{ iff } z \in \mathbb{R}$$

Corollary 16.4 Zeros of real polynomials come in complex conjugate pairs.

Theorem 16.5 Let $A \in \mathbb{R}^{n \times n}$ and let $\lambda = \lambda_1 \pm i\lambda_2$ be two eigenvalues. Then the corresponding e-vectors can be written as $\vec{v} = \vec{v}_1 \pm i\vec{v}_2$.

$$(\lambda_1 + i\lambda_2, \vec{v}_1 + i\vec{v}_2)$$

$$(\lambda_1 - i\lambda_2, \vec{v}_1 - i\vec{v}_2)$$

Exercise:

3) Find the complex e-values and e-vectors of the matrix

$$A = \begin{bmatrix} 2 & -1 \\ 5 & 0 \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 2-\lambda & -1 \\ 5 & -\lambda \end{vmatrix} = \lambda^2 - 2\lambda + 5$$

$$= (\lambda - 1)^2 + 4$$

$$(\lambda - 1)^2 + 4 = 0$$

$$(\lambda - 1)^2 = -4$$

$$\lambda - 1 = \pm 2i$$

$$\lambda = 1 \pm 2i$$

$$\lambda = 1 + 2i : A - \lambda I = \begin{bmatrix} 1-2i & -1 \\ 5 & -1-2i \end{bmatrix}$$

$$\vec{v} = \begin{bmatrix} 1+2i \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \end{bmatrix} + i \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$(1+2i, \begin{bmatrix} 1 \\ 5 \end{bmatrix} + i \begin{bmatrix} 2 \\ 0 \end{bmatrix})$$

$$(1-2i, \begin{bmatrix} 1 \\ 5 \end{bmatrix} - i \begin{bmatrix} 2 \\ 0 \end{bmatrix})$$

Matrix Exponential and Linear Systems

$$\text{Recall: } e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

Def 16.6 Let $A \in \mathbb{R}^{n \times n}$ and define the matrix exponential of A as the formal power series

$$e^{At} := \sum_{n=0}^{\infty} \frac{A^n t^n}{n!} = I + At + \frac{A^2 t^2}{2!} + \dots$$

Theorem 16.7 If $A = S \Lambda S^{-1}$, then

$$e^{At} = S e^{\Lambda t} S^{-1} = S \begin{bmatrix} e^{\lambda_1 t} & & \\ & \ddots & \\ & & e^{\lambda_n t} \end{bmatrix} S^{-1}$$

Proof: Set $A = S \Lambda S^{-1}$ and compute

$$e^{At} = \sum_{n=0}^{\infty} \frac{A^n t^n}{n!} = \sum_{n=0}^{\infty} \frac{S \Lambda^n S^{-1} t^n}{n!}$$

$$\begin{aligned} \sum_{k=0}^{\infty} \frac{\Lambda^k t^k}{k!} &= \sum_{k=0}^{\infty} \begin{bmatrix} \frac{\lambda_1^k t^k}{k!} & & 0 \\ & \ddots & \\ 0 & & \frac{\lambda_n^k t^k}{k!} \end{bmatrix} = S \left(\sum_{n=0}^{\infty} \frac{\Lambda^n t^n}{n!} \right) S^{-1} \\ &= \begin{bmatrix} \sum \frac{\lambda_1^k t^k}{k!} & & 0 \\ & \ddots & \\ 0 & & \sum \frac{\lambda_n^k t^k}{k!} \end{bmatrix} = S e^{\Lambda t} S^{-1} \end{aligned}$$

Now for $\Lambda = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$ we have

$$\text{that } e^{\Lambda t} = \begin{bmatrix} e^{\lambda_1 t} & & \\ & \ddots & \\ & & e^{\lambda_n t} \end{bmatrix}.$$

Properties 16.8

- i. If $AB = BA$, then $e^{At} e^{Bt} = e^{(A+B)t}$
- ii. e^{At} is invertible with inverse $(e^{At})^{-1} = e^{-At}$

$$\text{iii. } \frac{d}{dt} e^{At} = Ae^{At}$$